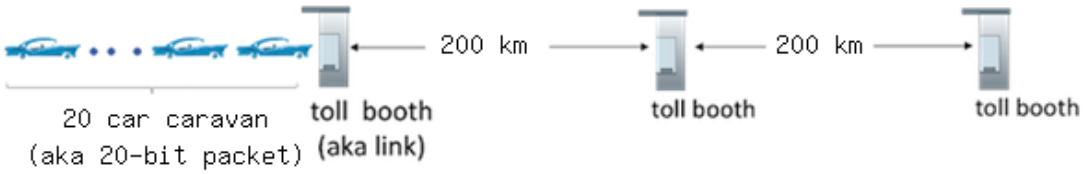
HOMEWORK 1

1. Write the equivalent of 5Gbits in Kbits, Megabits and bits. Show the conversions clearly.
2. Show the connection between Gigabyte, gigabits, Kilobyte, kilobits and bits.
3. Write equivalent of 5ms in seconds, microseconds. Show the conversions clearly.
4. Let us consider how long it takes to send a file of 640,000 bits from Host A to Host B over a circuit-switched network. Suppose that all links in the network use TDM with 24 slots and have a bit rate of 1.536 Mbps. Also suppose that it takes 500 msec to establish an end-to-end circuit before Host A can begin to transmit the file.
    a. How long does it take to send the file?
    b. What will the file transmission time be if packet switching is used? Let's assume that the entire file can be sent immediately once packet transmission starts. We can also assume no significant propagation delay or queuing delay in this simplified scenario.
    c. Let's recalculate (b) by taking out the assumptions made. Let's assume that the file is transmitted in packets of 1,500 bytes, including the headers. Let's also assume that propagation delay from Host A to Host B is 5 milliseconds per packet, and processing and queuing delays are negligible.
5. Consider the figure below, which draws the analogy between store-and-forward link transmission and propagation of bits in packet along a link, and cars in a caravan being serviced at a toll booth and then driving along a road to the next tollbooth.

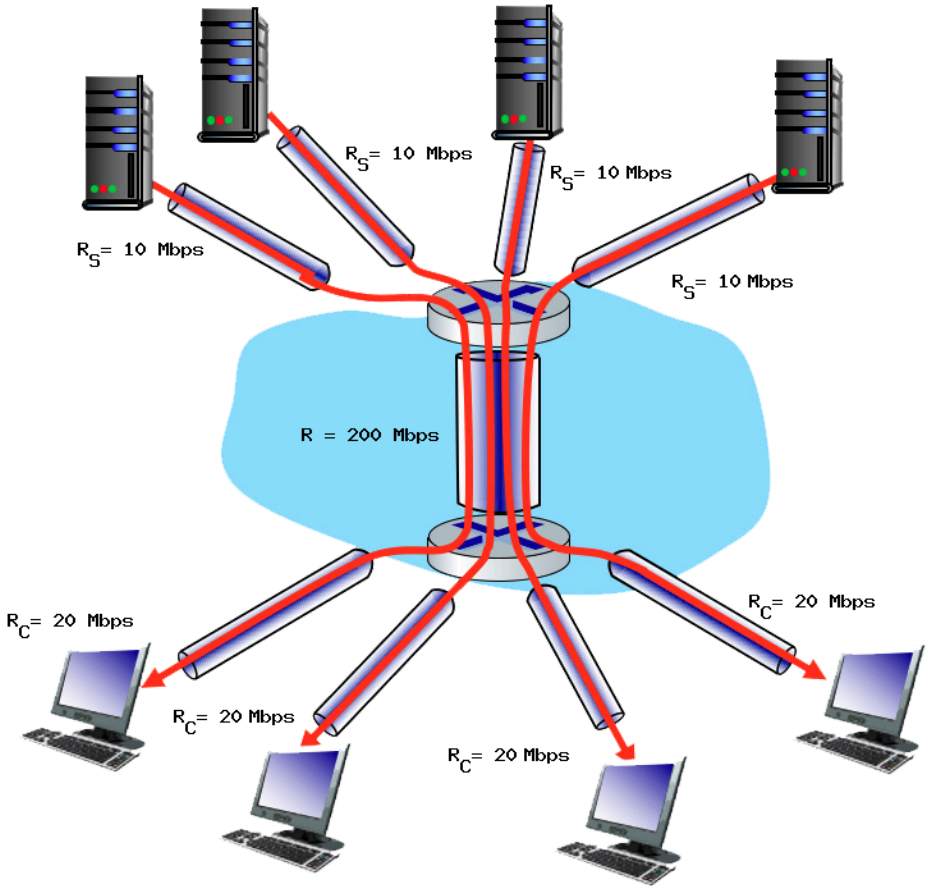
    Suppose the caravan has 20 cars, and that the tollbooth services (that is, transmits) a car at a rate of one car per 5 seconds. Once receiving serving a car proceeds to the next tool both, which is 200 kilometres away at a rate of 10 kilometres per second. Also assume that whenever the first car of the caravan arrives at a tollbooth, it must wait at the entrance to the tollbooth until all of the other cars in its caravan have arrived, and lined up behind it before being serviced at the toll booth. (That is, the entire caravan must be stored at the tollbooth before the first car in the caravan can pay its toll and begin driving towards the next tollbooth).



a. Once a car enters service at the tollbooth, how long does it take until it leaves service?
b. How long does it take for the entire caravan to receive service at the tollbooth (that is the time from when the first car enters service until the last car leaves the tollbooth)?
c. Once the first car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?
d. Once the last car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?
e. Once the first car leaves the tollbooth, how long does it take until it enters service at the next tollbooth?
f. Are there ever two cars in service at the same time, one at the first toll booth and one at the second toll booth? Answer Yes or No.
g. Are there ever zero cars in service at the same time, i.e., the caravan of cars has finished at the first toll both but not yet arrived at the second tollbooth? Answer Yes or No.

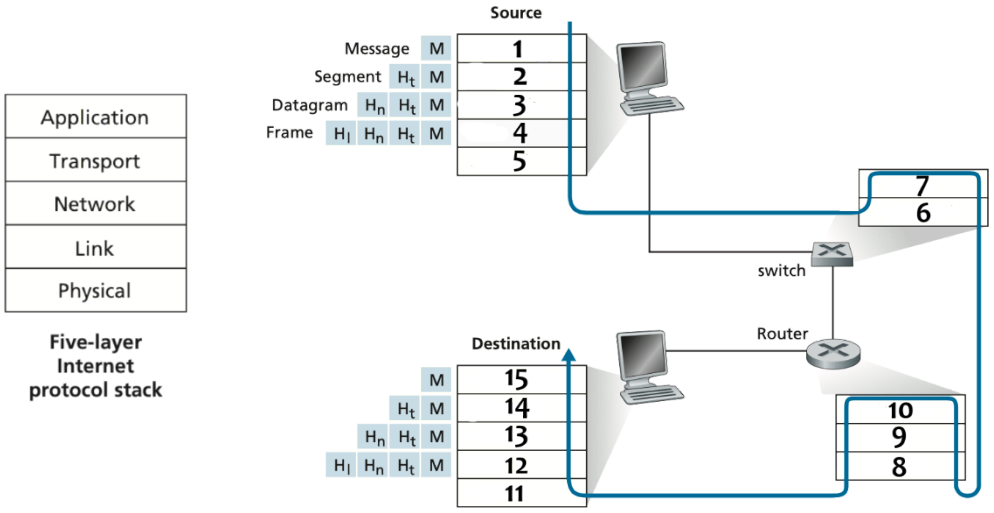
6. Consider this scenario, in which a single router is transmitting packets, each of length L bits, over a single link with transmission rate R Mbps to another router at the other end of the link. Suppose that the packet length is L= 16000 bits, and that the link transmission rate along the link to router on the right is R = 10 Mbps.
    a. What is the transmission delay?
    b. What is the maximum number of packets per second that can be transmitted by this link?
7. Consider the scenario where four different servers are connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of R = 200 Mbps. The four links from the servers to the shared link have a

transmission capacity of R_S = 10 Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of R_C = 20 Mbps.



a. What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?
b. Which link is the bottleneck link? Format as Rc, Rs, or R.
c. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (R_S)? Answer as a decimal.
d. Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (R_C)? Answer as a decimal.
e. Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal.

8. In the scenario below, imagine that you're sending an http request to another machine somewhere on the network.



a. What layer in the IP stack best corresponds to the phrase: *'passes frames from one node to another across some medium*'.
b. What layer in the IP stack best corresponds to the phrase: *'bits live on the wire'*.
c. What layer in the IP stack best corresponds to the phrase: *'moves datagrams from the source host to the destination host'*.
d. What layer in the IP stack best corresponds to the phrase: 'handles messages from a variety of network applications'.
e. What layer in the IP stack best corresponds to the phrase: 'handles the delivery of segments from the application layer, may be reliable or unreliable'.
f. What layer corresponds to box 1?
g. What layer corresponds to box 2?
h. What layer corresponds to box 3?
i. What layer corresponds to box 4?
j. What layer corresponds to box 5?
k. What layer corresponds to box 6?
l. What layer corresponds to box 7?
m. What layer corresponds to box 8?
n. What layer corresponds to box 9?

o. What layer corresponds to box 10?
p. What layer corresponds to box 11?
q. What layer corresponds to box 12?
r. What layer corresponds to box 13?
s. What layer corresponds to box 14?
t. What layer corresponds to box 15?